Daño Pulmonar

- relacionado al tabaco y derivados -

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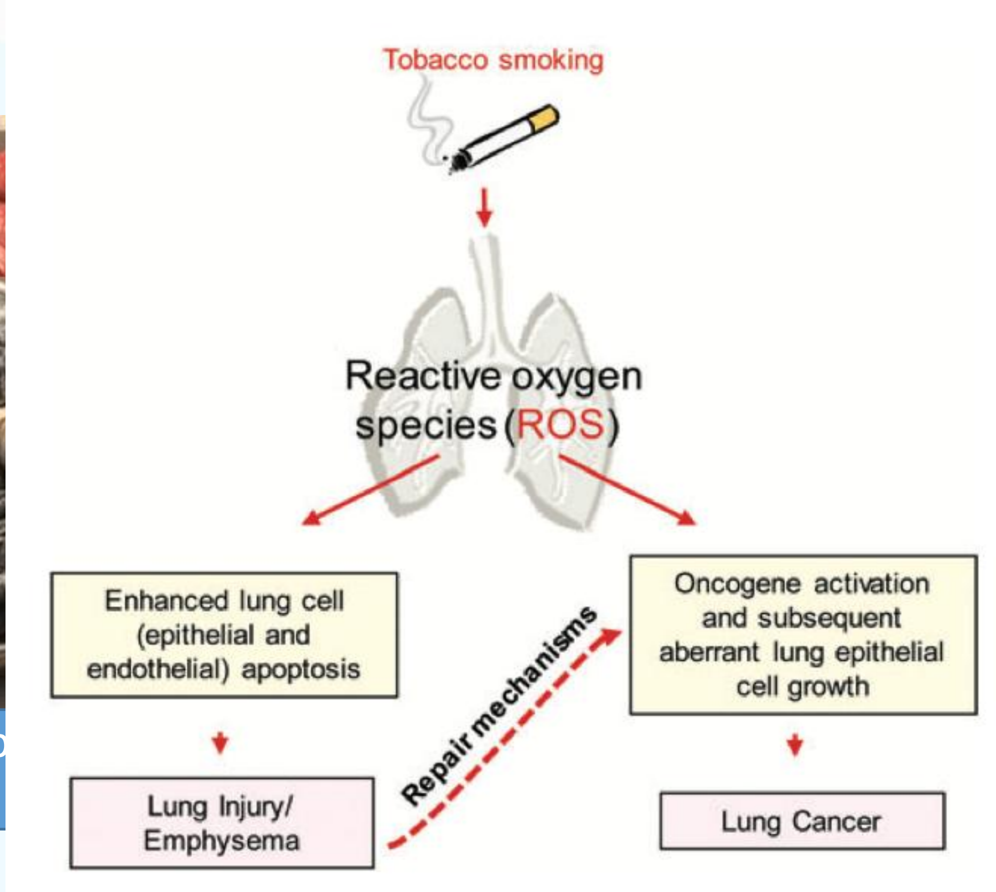
¿Que se nos viene a la mente cuando pensamos en daño pulmonar por cigarrillo?

Enfermedad Pulmonar Ob
(EPOC)

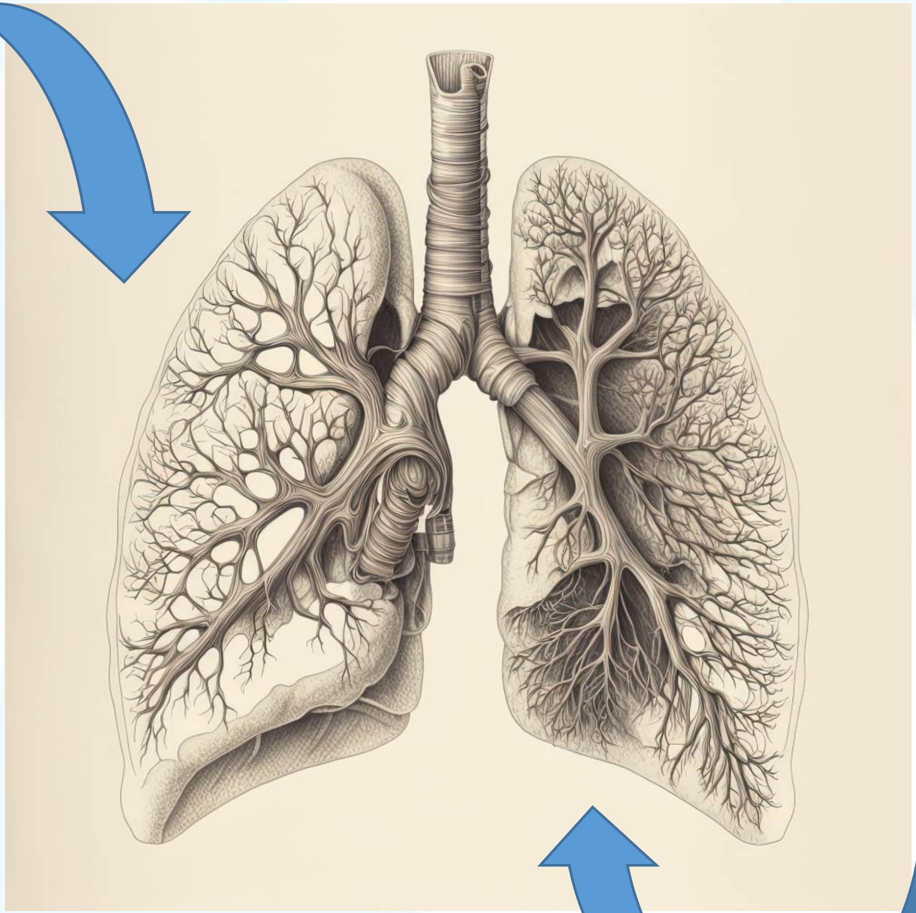


er de Pulmón

ANTIOXIDANTS & REDOX SIGNALING. Volume 21, Number 15, 2014 a Mary Ann Liebert, Inc. DOI: 10.1089/ars.2013.5469

Repercusiones del daño pulmonar por tabaco y productos relacionados

- Irritación y tos
- Hiperreactividad bronquial, tos y disnea
- Bronquitis
- EVALI



- smoking related-interstitial lung diseases (SR-ILD)
  - respiratory bronchiolitis (RB)
  - respiratory bronchiolitis-ILD (RB-ILD)
  - desquamative interstitial pneumonitis (DIP)
  - pulmonary Langerhans cell histiocytosis (PLCH)
- other
  - acute eosinophilic pneumonia (AEP)
  - combined pulmonary fibrosis and emphysema (CPFE)
- neoplasms
  - lung cancer
  - tracheal tumors
- chronic obstructive pulmonary disease
  - emphysema
  - constrictive bronchiolitis
- Quistes y bullas.
- Infecciosas en pleura y pulmón

Corto plazo → Mediano y largo plazo

Tiempo e intensidad de exposición a tabaco y derivados

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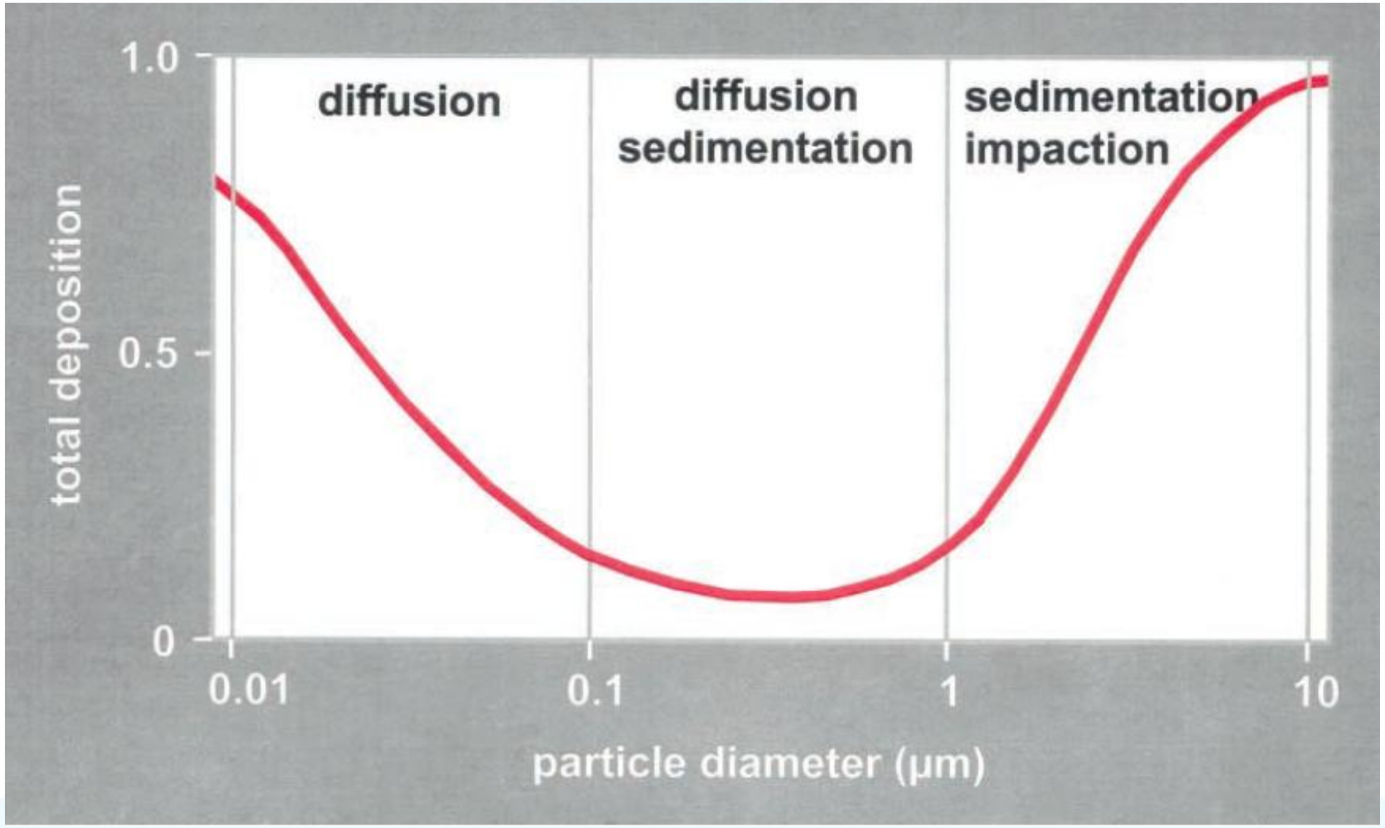
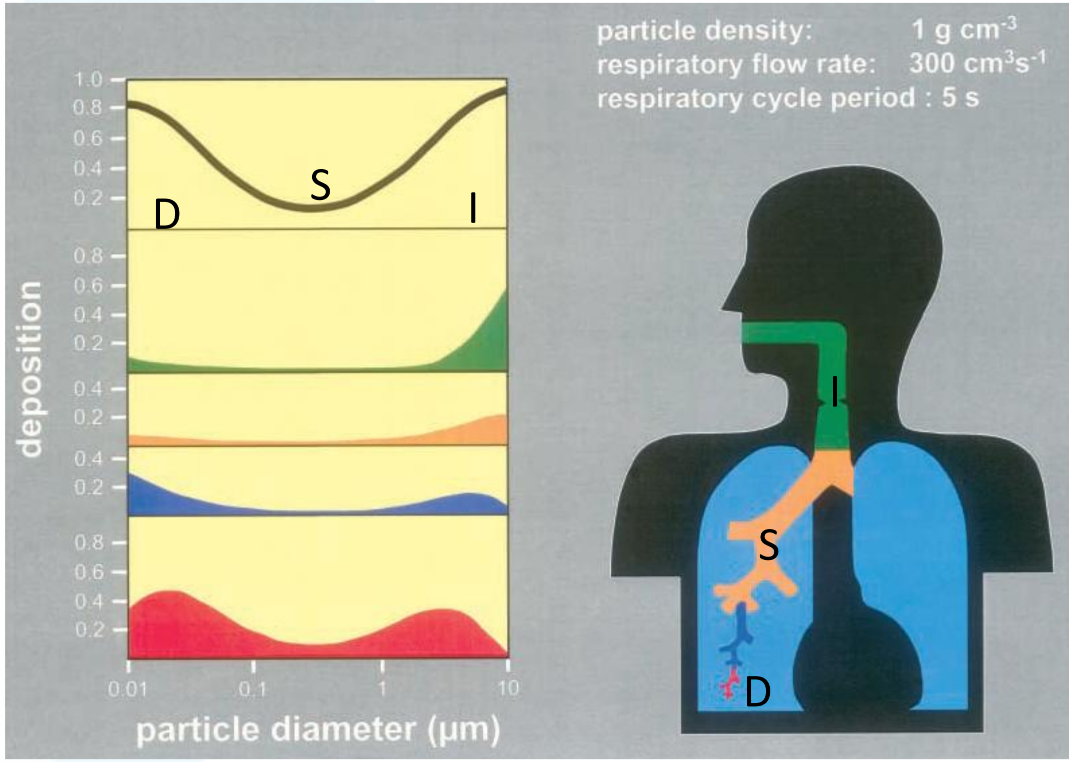
Emisión de gases a partir de cigarrillos

Deposición de partículas en el árbol respiratorio de acuerdo al tamaño





Heyder J. Proc Am Thorac Soc Vol 1. pp 315–320, 2004

Partículas del humo de cigarrillo en el pulmón humano

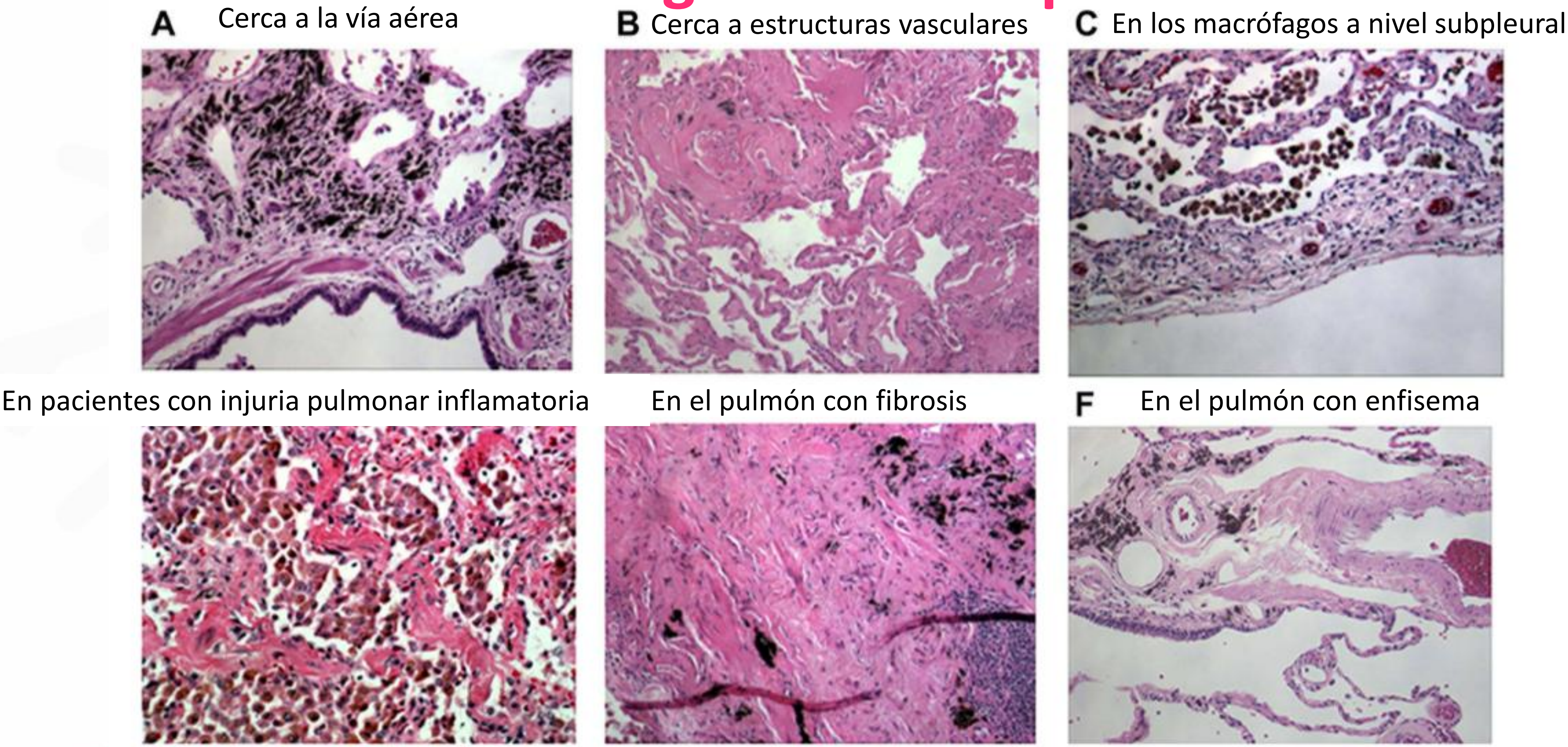
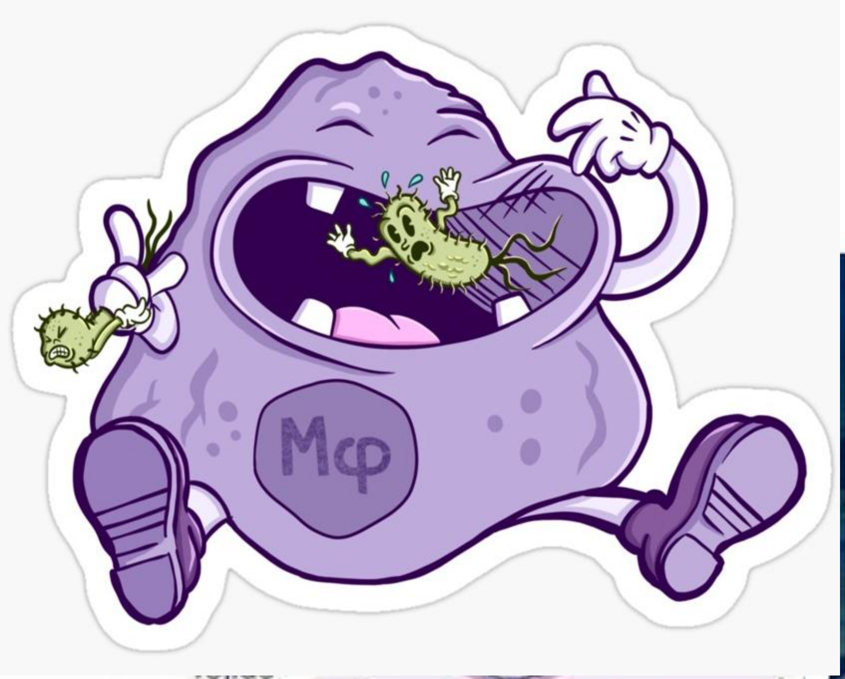
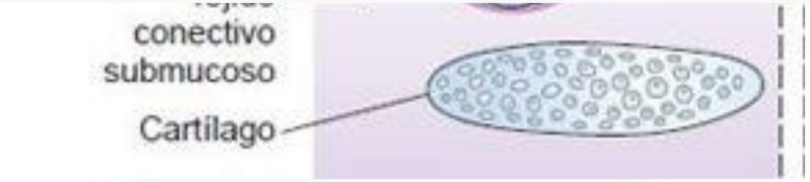


Figure 3 CSP in the human lung. Smokers show particle in close proximity to both airways (**A**) and vascular structures (**B**) and intracellularly within macrophages in distant sub-pleural regions (**C**). CSP is evident in lung resected from patients with inflammatory lung injury (**D**), fibrosis (**E**), and emphysema (**F**). Stain is hematoxylin and eosin. Magnification approximates 100×.

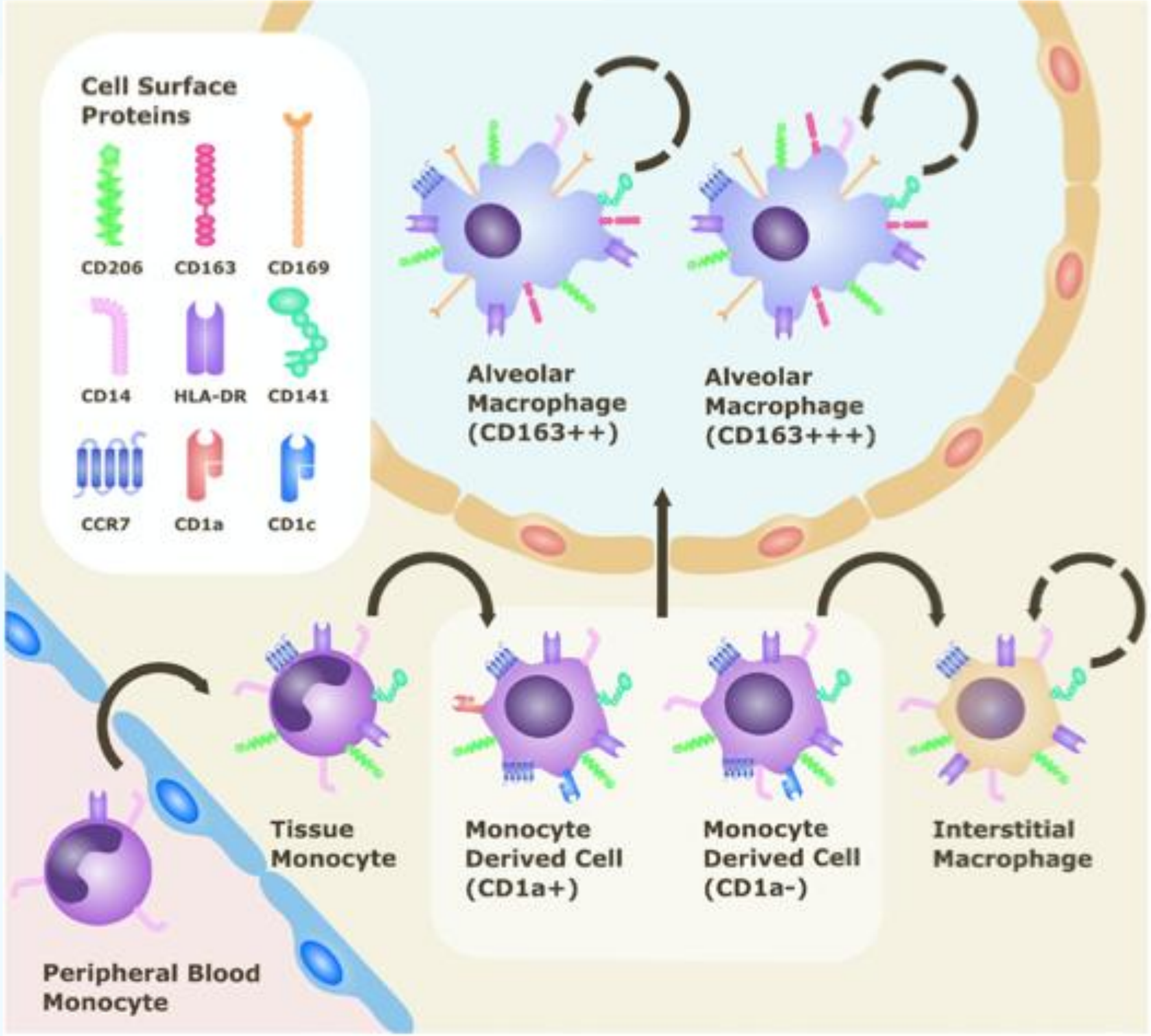
International Journal of Chronic Obstructive Pulmonary Disease 2022:17 117–140

Entonces, hay efecto citotóxico directo e instauración del proceso inflamatorio…





Marcadores de superficie de las diferentes poblaciones de macrófagos /monocitos en el pulmón



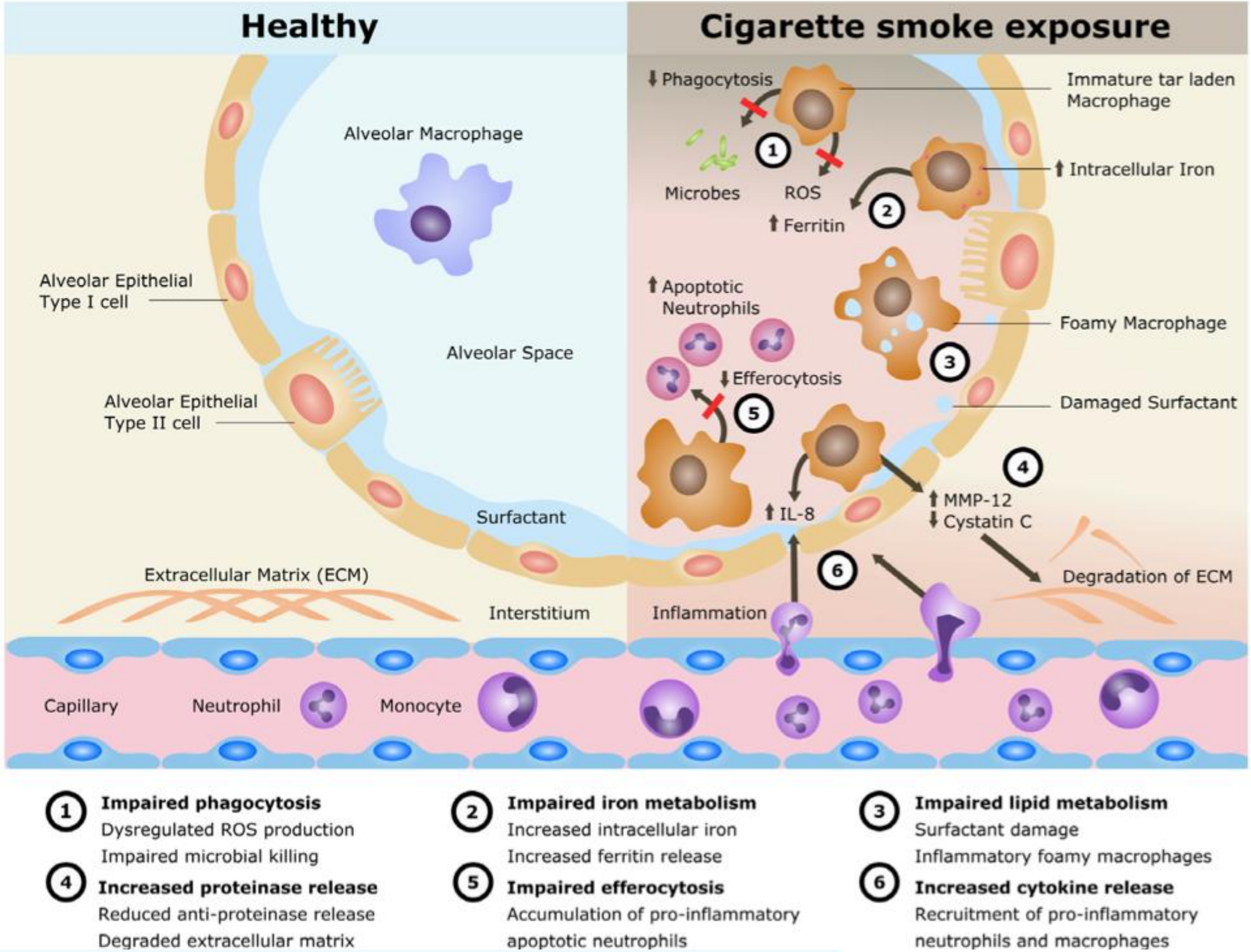
Lugg ST, *et al*. *Thorax* 2022;**77**:94–101. doi:10.1136/thoraxjnl-2020-216296

Cambios en la expresión de marcadores de superficie de los macrófagos alveolares por efecto del cigarrillo

Table 1 Cigarette smoking-related phenotypic changes of alveolar macrophages

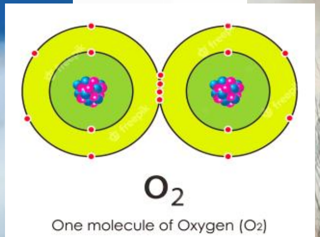
Cell surface marker	Function	Level of expression in smokers
CD11c	Integrin alpha X chain; adherence to stimulated endothelium and phagocytosis of complement-coated particles	↑
CD14	Co-receptor working with toll-like receptors; facilitates response to lipopolysaccharide	↑
CD16	Type III Fc gamma binds immunoglobulin; participates in signal transduction and phagocytosis	↓
CD31	Platelet endothelial cell adhesion molecule; promotes tethering of apoptotic cells in efferocytosis	↓
CD36	Scavenger receptor; regulates efferocytosis of apoptotic neutrophils and role in fatty acid uptake and lipid metabolism	↓
CD44	Cell surface glycoprotein involved in cell–cell interactions, adhesion and migration; receptor of efferocytosis	↓
CD54	Intercellular adhesion molecule-1; involved in cell–cell interactions and leucocyte migration	↓
CD71	Transferrin receptor; role in iron metabolism and iron import into cells through endocytosis	↓
CD91	Low density lipoprotein receptor-related protein; binds to collagen like region on lung collectins and role in phagocytosis	↓
CD163	Scavenger receptor for haemoglobin–haptoglobin complex; innate immune receptor for bacteria and can be influenced by corticosteroids	↓

Lugg ST, *et al*. *Thorax* 2022;**77**:94–101. doi:10.1136/thoraxjnl-2020-216296

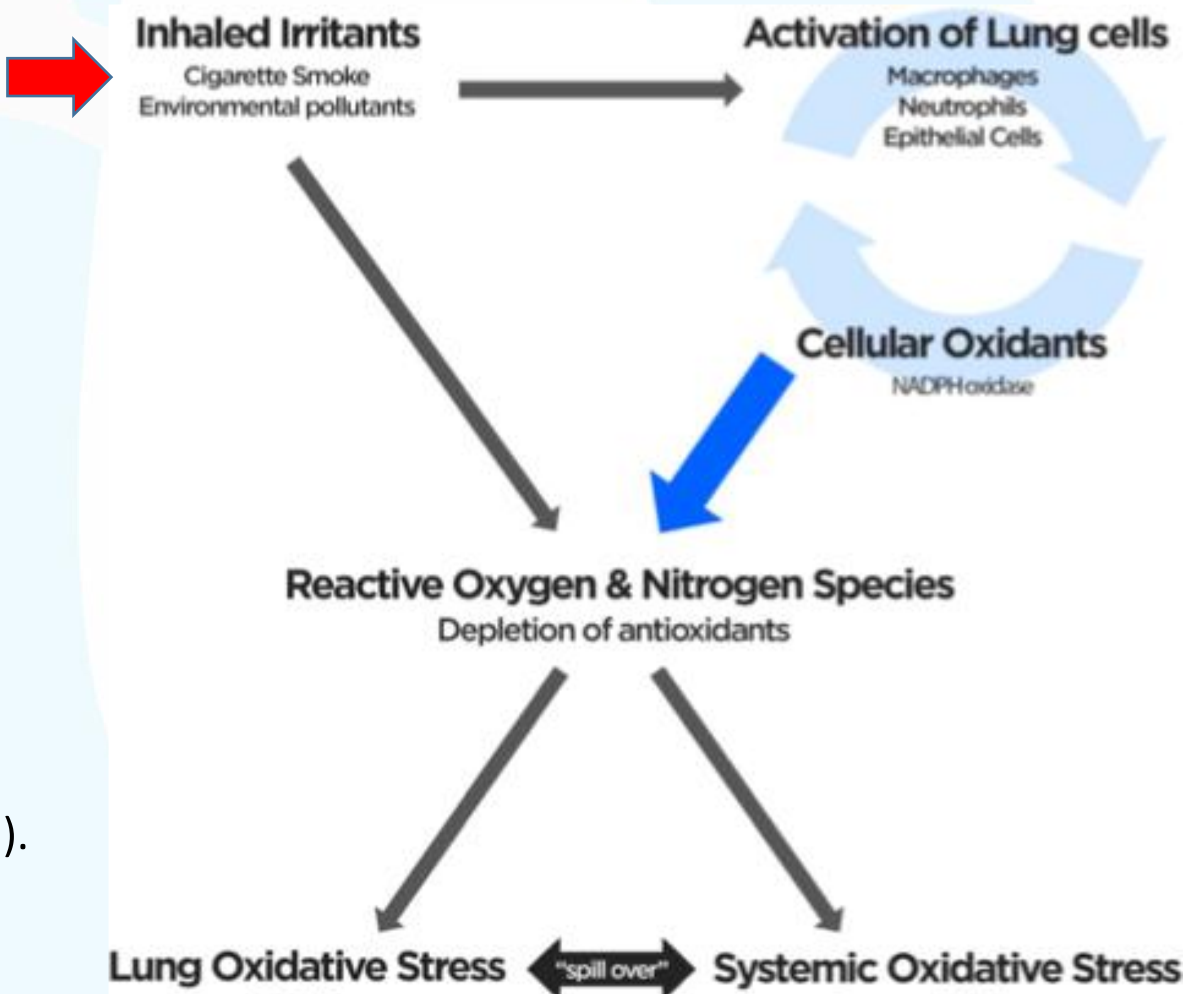


Lugg ST, *et al*. *Thorax* 2022;**77**:94–101. doi:10.1136/thoraxjnl-2020-216296

Oxigeno: Beneficio y riesgo



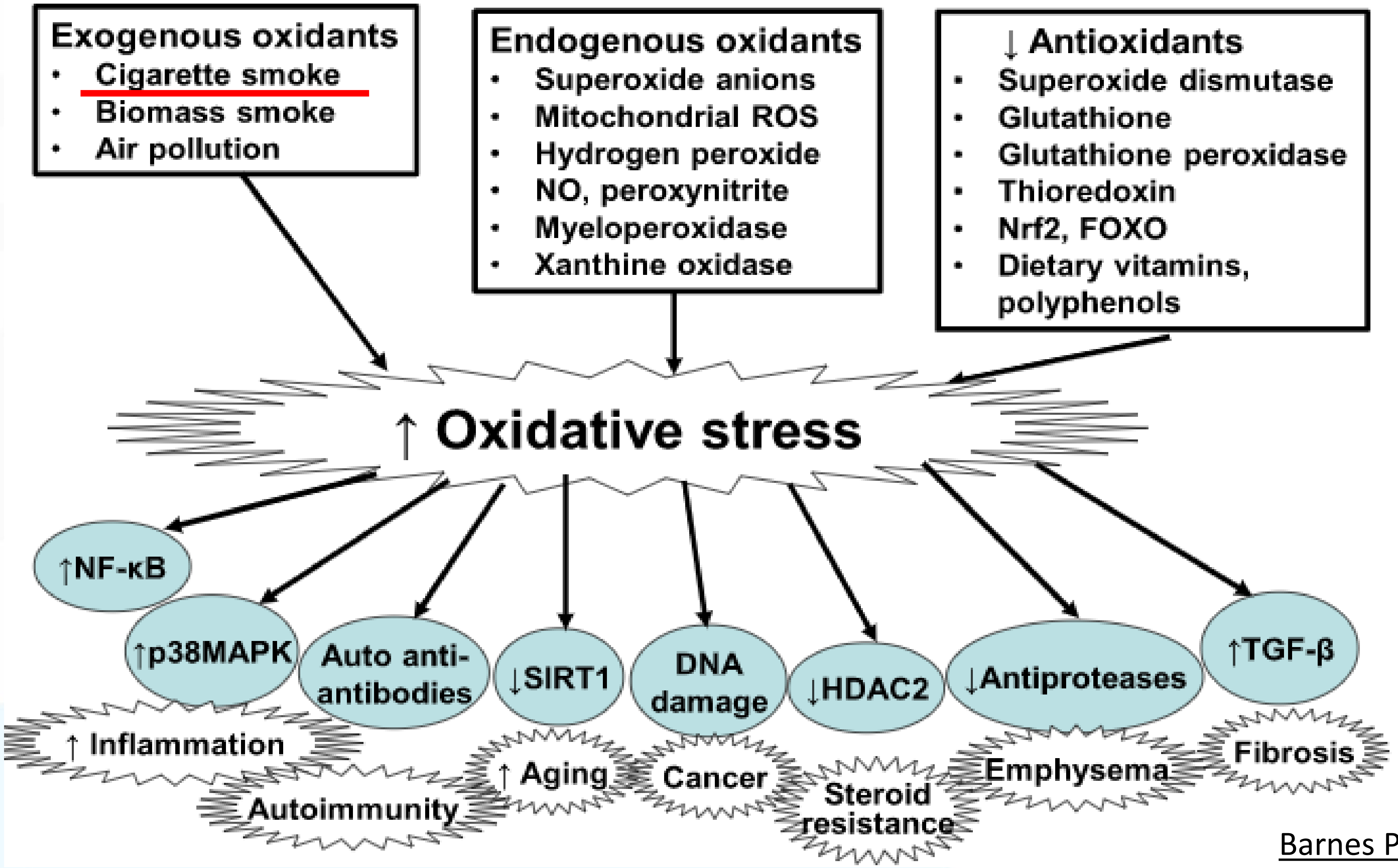
Estrés oxidativo en las enfermedades Respiratorias



EPOC, asma, infecciones, FPI, bronquiectasias, SDRA

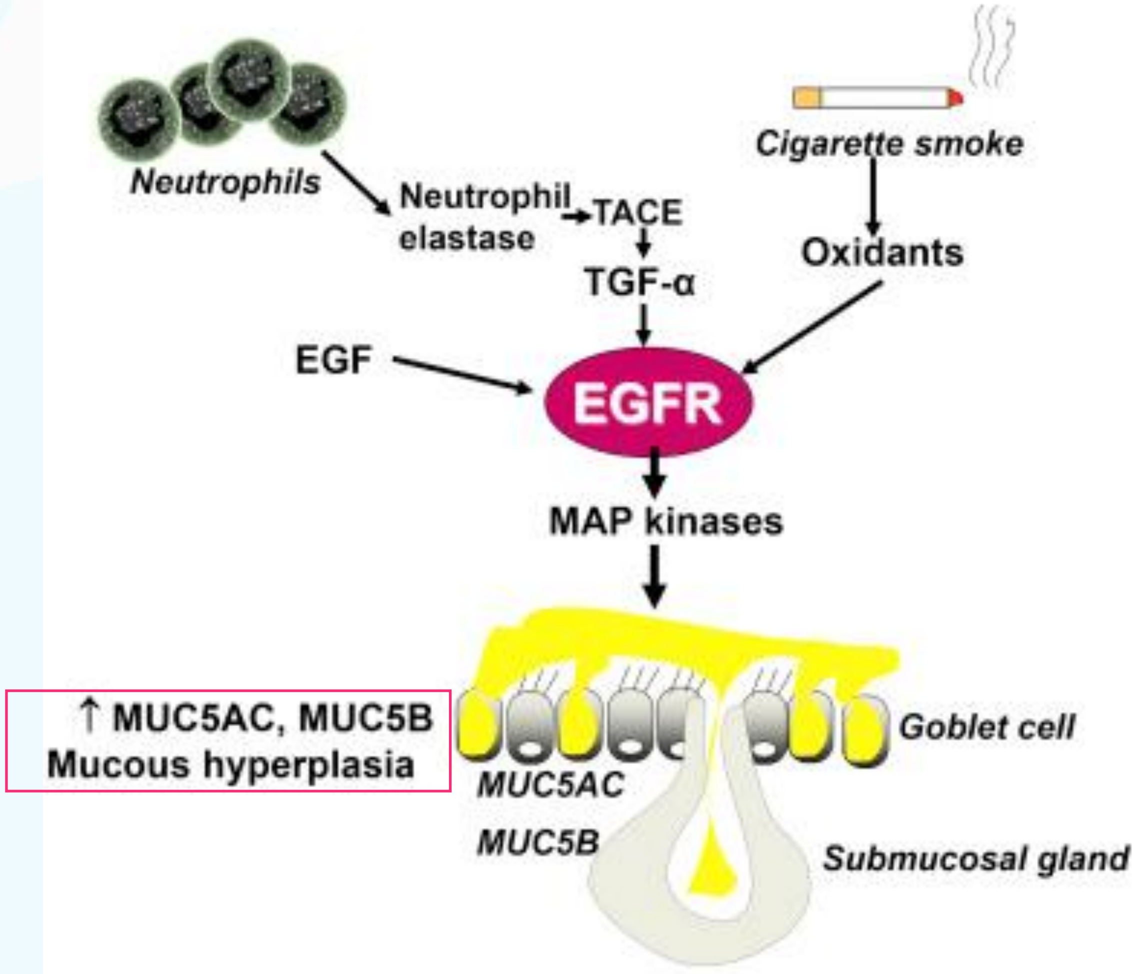
1. Rahman, I. Current Opinion in Pharmacology 12, 256–265 (2012).
2. Bernardo I. Pharmacology & Therapeutics 155, 60–79 (2015).
3. Dua, K. et al. Chemico-Biological Interactions 299, 168–178 (2019).

Repercusiones del estrés oxidativo en el pulmón



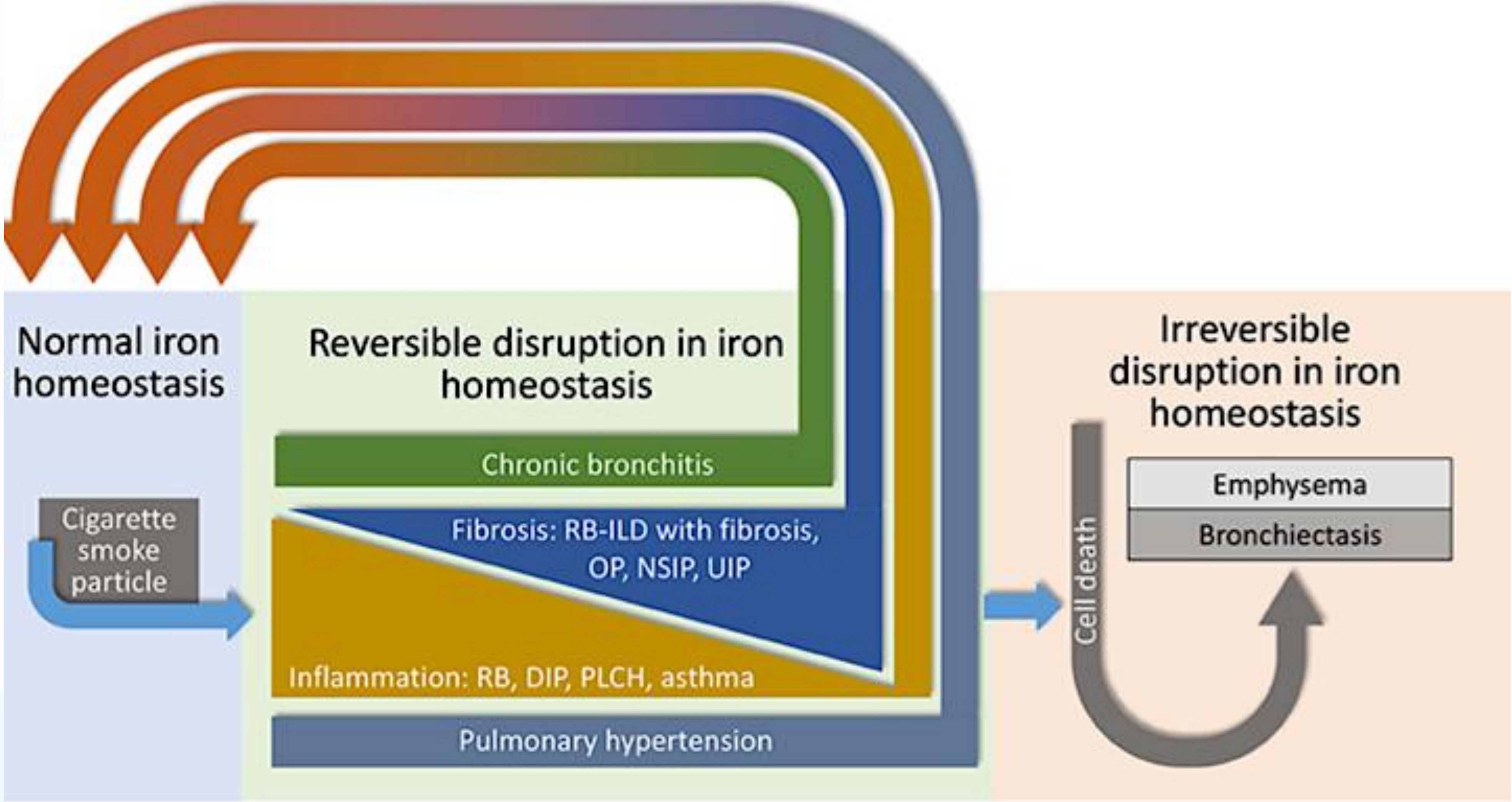
Barnes P. .redox.2020.101544

El estrés oxidativo por cigarrillo, relacionado a mayor producción de moco



https://doi.org/10.1016/j.jaci.2016.05.011

Partículas del humo de cigarrillo y relación con disrupción de la hemostasis del hierro que deriva en lesión pulmonar (hipótesis!!!)



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Genotoxicidad

Tipos de carcinógenos en los productos del tabaco

Tobacco smoke

Chemical class	*Number of compounds*[‡]	*Representative carcinogens and typical amounts in mainstream smoke (ng per cigarette)*[§]	
PAH[‖]	14	BaP	9
		Dibenz[*a,h*]anthracene	4
Nitrosamines	8	NNK	123
		NNN	179
Aromatic amines[¶]	12	4-Aminobiphenyl	1.4
		2-Naphthylamine	10
Aldehydes	2	Formaldehyde	16,000
		Acetaldehyde	819,000
Phenols	2	Catechol	68,000
Volatile hydrocarbons	3	Benzene 1,3-Butadiene	59,000 52,000
Nitro compounds	3	Nitromethane	500
Other organic compounds	8	Ethylene oxide Acrylonitrile	7,000 10,000
Inorganic compounds	9	Cadmium	132
Total	**61**		

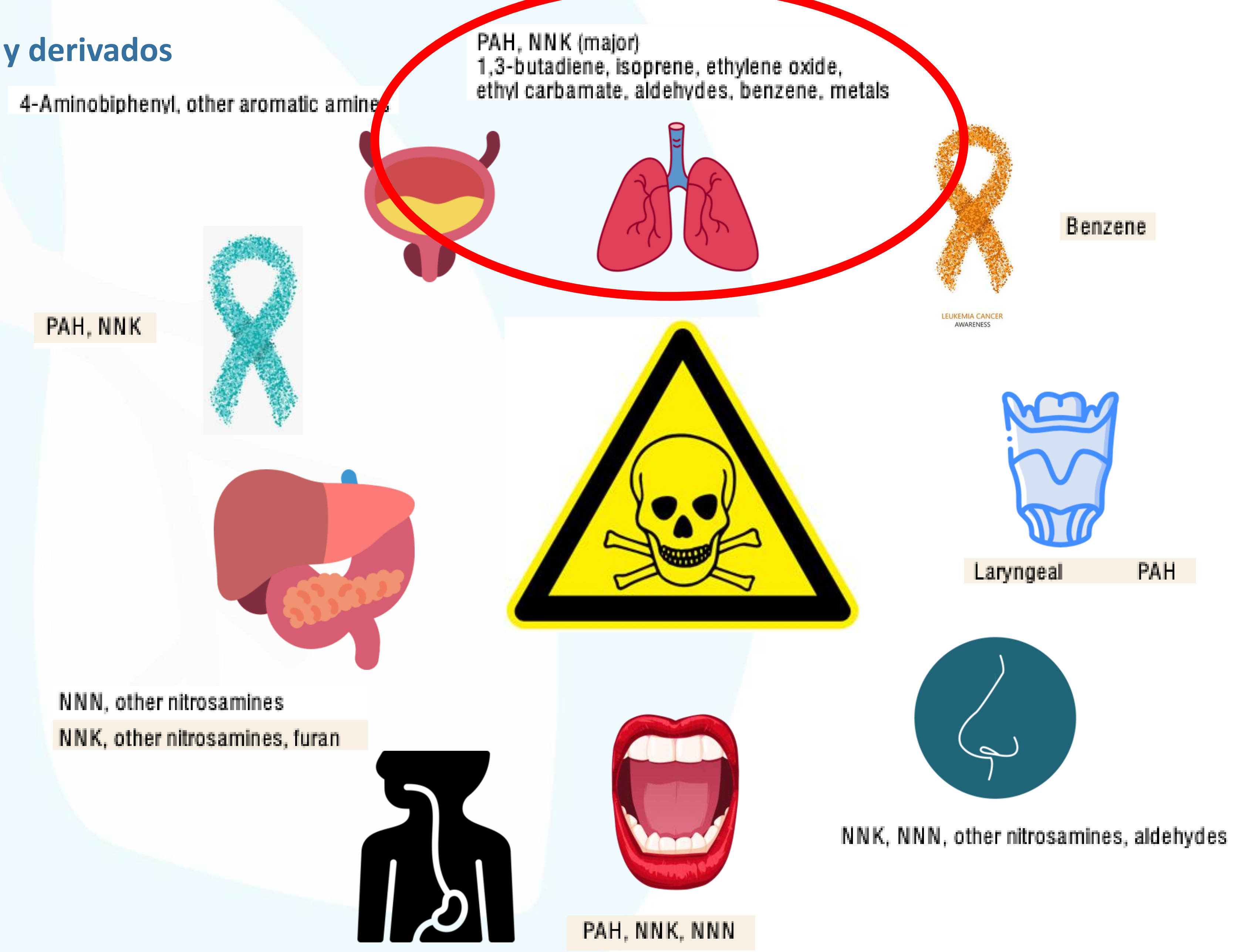
Unburned tobacco

Chemical class	*Number of compounds*	*Representative carcinogens and typical amounts in processed tobacco (ng g^{-1})*	
PAH	1	BaP	0.4–90
Nitrosamines	6	NNK	1,890
		NNN	8,730
Aldehydes	2	Formaldehyde	1,600–7,400
		Acetaldehyde	1,400–7,400
Inorganic compounds	7	Cadmium	1,300–1,6000
Total	**16**		

*Adapted from REFS 3,6. This table shows only carcinogens that have been evaluated by the International Agency for Research on Cancer (IARC). Others might also be present. [‡]Evaluated by the IARC; sufficient evidence for carcinogenicity in either laboratory animals or humans. Some of these carcinogens (two PAHs, three heterocyclic PAH analogues and four nitrosamines) have been detected only sporadically in smoke and are often below the detection limits of conventional assays. [§]From REFS 3,6,10,11. [‖]Including heterocyclic analogues. BaP, benzo[*a*]pyrene; NNK, 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone; NNN, *N'*-nitrosonornicotine; PAH, polycyclic aromatic hydrocarbons.

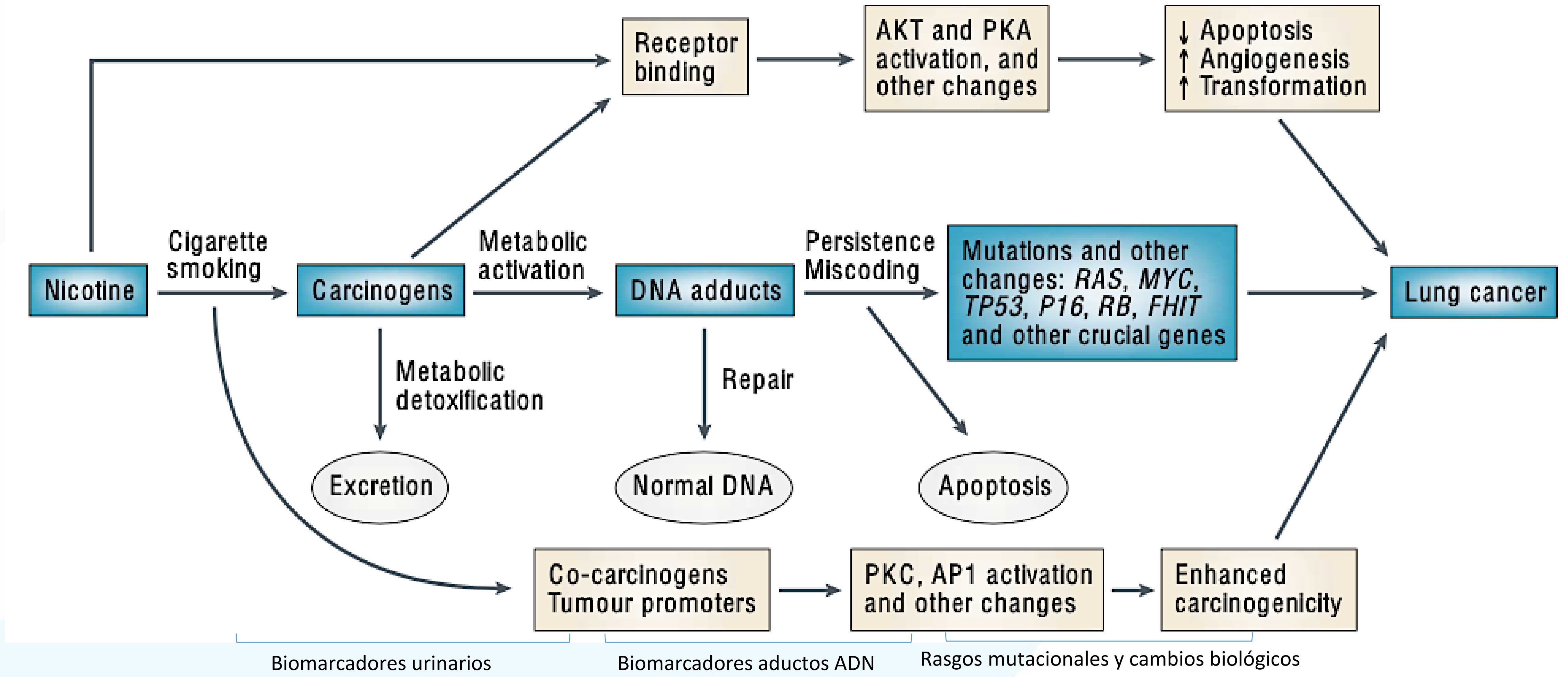
Hecht, S. Tobacco carcinogens, their biomarkers and tobacco-induced cancer. *Nat Rev Cancer* **3,** 733–744 (2003). https://doi.org/10.1038/nrc1190

Carcinogenicidad multisistémica del tabaco.



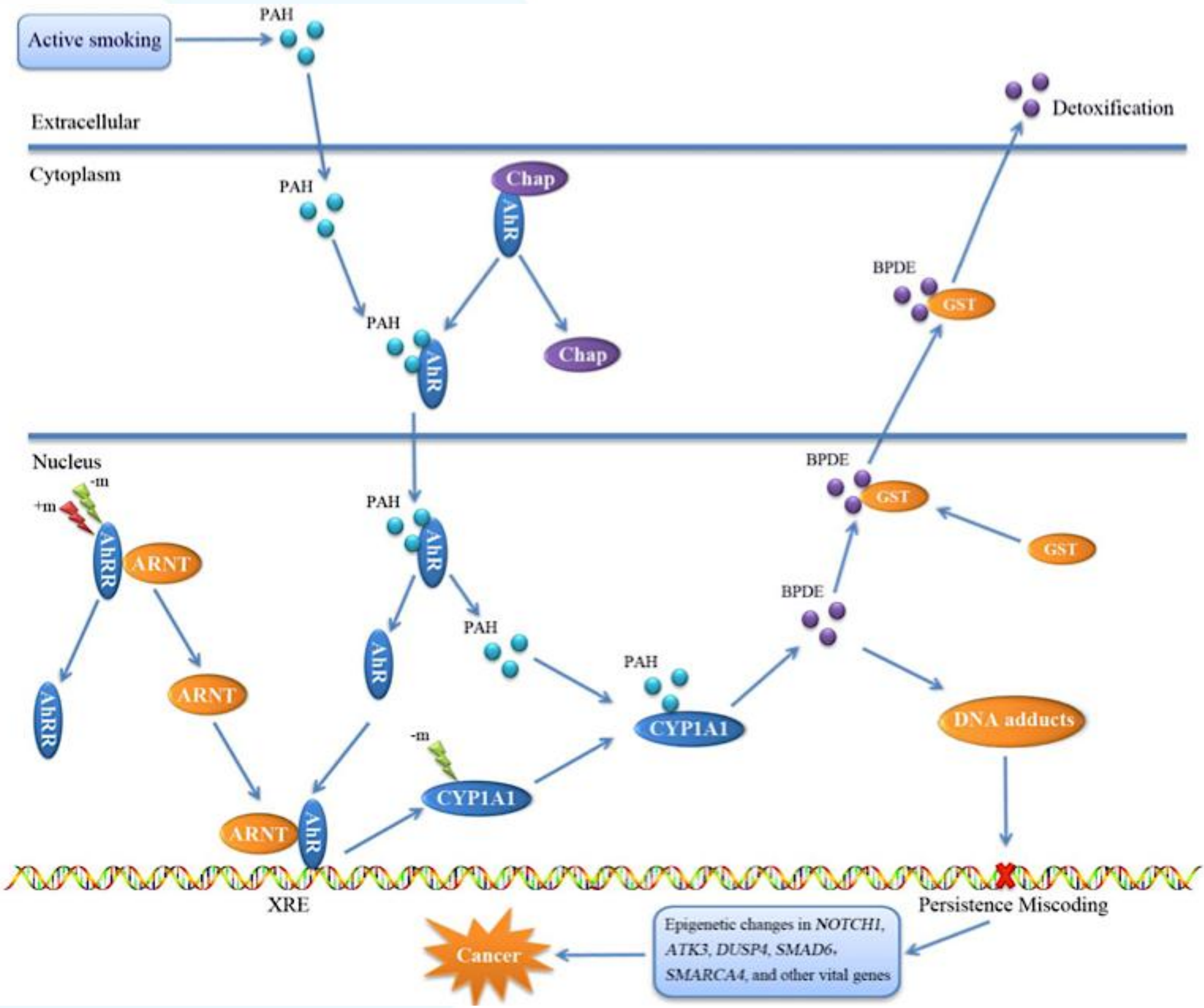
Hecht, S. Tobacco carcinogens, their biomarkers and tobacco-induced cancer. *Nat Rev Cancer* **3,** 733–744 (2003). https://doi.org/10.1038/nrc1190

Asociación entre la adicción a la nicotina y cáncer de pulmón a través de carcinógenos en el humo de cigarrillo



Hecht, S. Tobacco carcinogens, their biomarkers and tobacco-induced cancer. *Nat Rev Cancer* **3,** 733–744 (2003). https://doi.org/10.1038/nrc1190

Formación de aductos de ADN en una de las vías hiper-expresadas

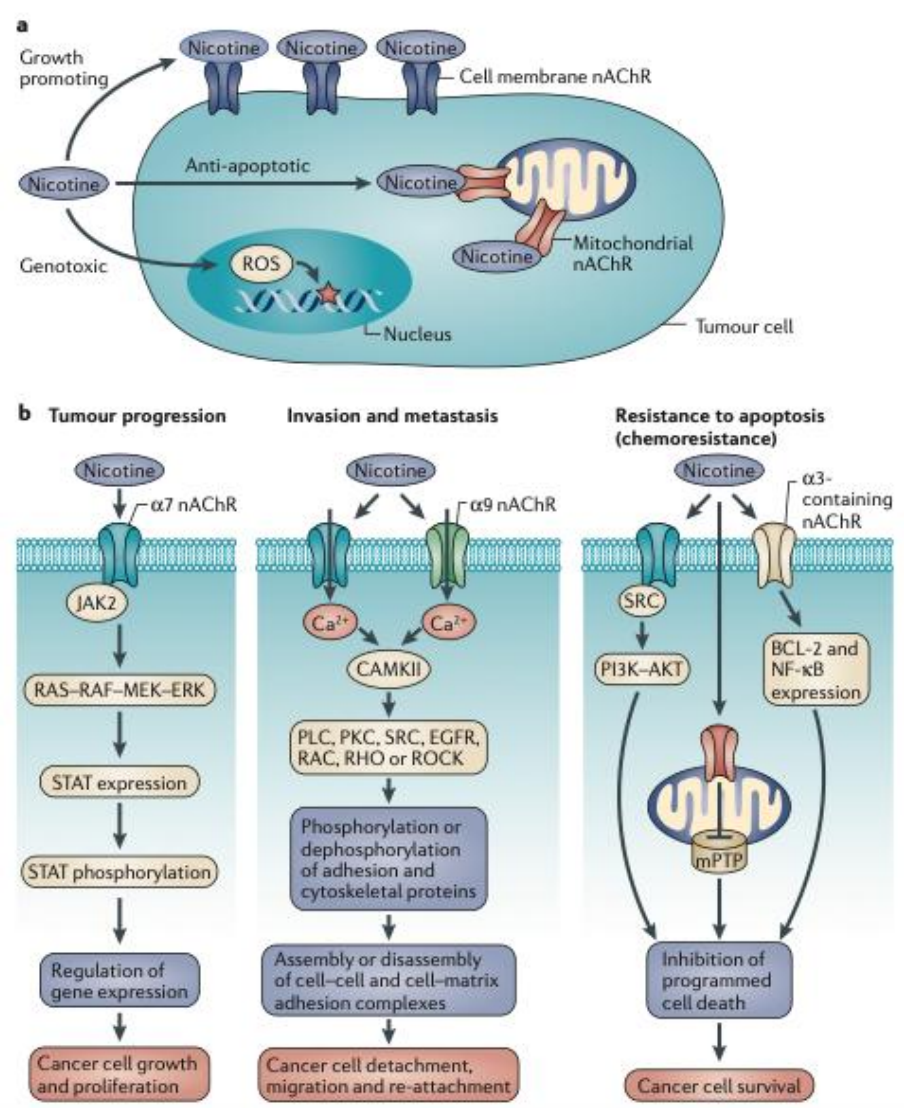


Ma, Y., Li, M.D. *Sci Rep* **7,** 1811 (2017). https://doi.org/10.1038/s41598-017-01856-4

Daño Pulmonar relacionado al tabaco y derivados

La nicotina como carcinógeno y efectos sobre las células tumorales en el cáncer establecido

"La data acumulada en la literatura indica que la nicotina tiene la capacidad de dañar el genoma, los procesos metabólicos celulares, amplificar oncogenes , inactivar genes supresores y promover el microambiente de cancer."



Grando, S. Connections of nicotine to cancer. *Nat Rev Cancer* **14,** 419–429 (2014). https://doi.org/10.1038/nrc3725

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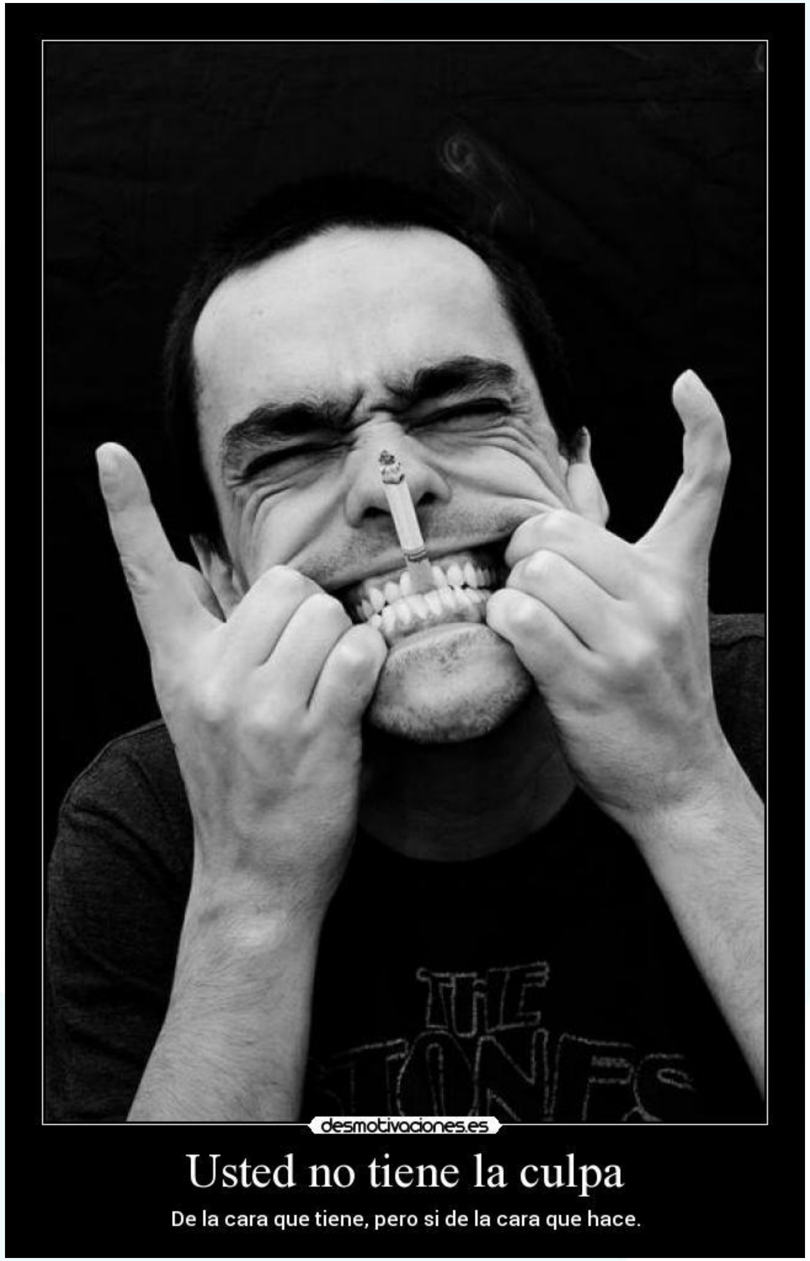

TIEMPO E INTENSIDAD

IPA: Nº CIG AL DIA /20 X AÑOS FUMANDO (>20, >30)

BRINKMAN: NUMERO DE CIG AL DIA POR AÑOS FUMANDO (300, 400?)

GOLD COPD 2024


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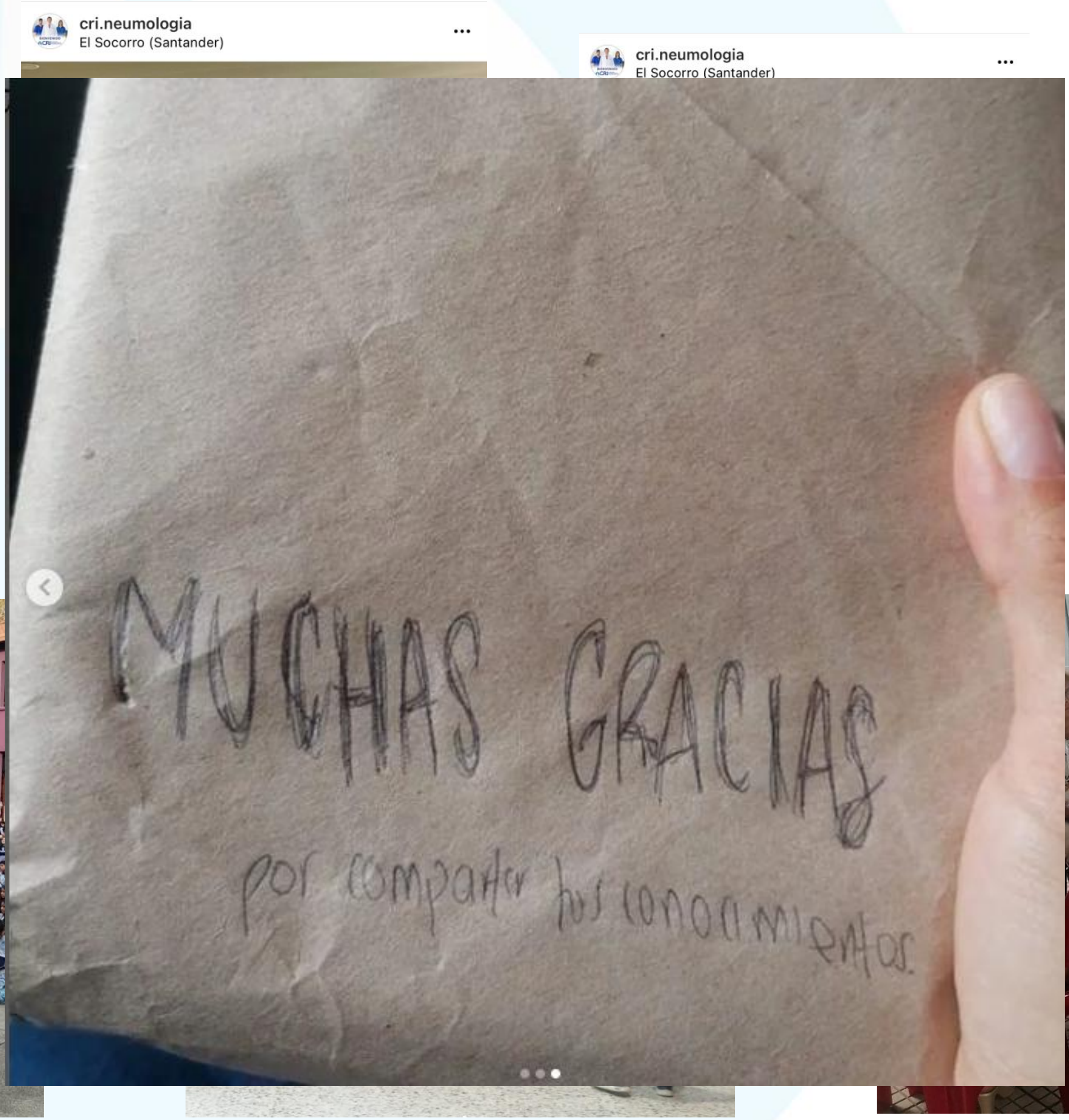
El Socorro (Santander)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